

SIGNS & SYSTEMS

FINAL LABORATORY PROJECT



2021

1. The code below plots the function

```

step=0.001;
t1 = -8:step:-4-step;
x1 = zeros(size(t1));
t2 = -4:step:3-step;
x2 = t2+2;
t3 = 3:step:8;
x3 = t3-2;
t = [t1 t2 t3];
x = [x1 x2 x3];
plot(t,x)

```

$$x(t) = \begin{cases} 0 & \text{if } t < -4 \\ t+2 & \text{if } -4 \leq t < 3 \\ t-2 & \text{if } 3 \leq t \end{cases}$$

Write the code to calculate the function

$$x[n] = \begin{cases} 0 & \text{if } n < 2 \\ 2n-4 & \text{if } 2 \leq n < 4 \\ 4-n & \text{if } 4 \leq n \end{cases}$$

2. Draw the following analog signal:

For $t \leq -1$, $x(t) = 0$

For $-1 < t \leq 0$, $x(t) = \cos(\pi t/2)$

For $0 < t \leq 1$, $x(t) = \exp(-t)$

For $1 < t$, $x(t) = 0$

3. Write code to calculate the convolution of the continuous-time functions $h(t)$ and $x(t)$.

$$h(t) = [2te^{-t} + e^{-2t} - e^{-3t}] \cdot u(t)$$

$$x(t) = [1 - e^{-1.5t}] \cdot u(t)$$

4. *What is the destination of the following program?*

```
% CONVOLUTION USING CONV (numerical convolution) % To calculate n , note
that  $y(40) = y(20 \times 2)$  \ n=0:20; kf = 0.1; m = 1; T = 2; h = (1-exp(-kf*n*T/m))/kf; x =
[ones(1,5)
-ones(1,5) zeros(1,10)]; y
= conv(T*h,x);

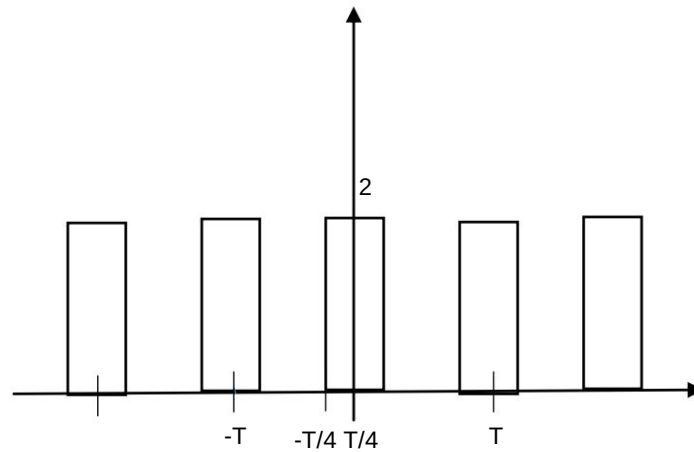
% EXACT SOLUTION (exact solution) t1 =
0:.1:9.9; t2 =
10:.1:19.9; t3 =
20:.1:40; ya =
[100*(0.1*t1-1+exp(-0.1*t1)),-100*(0.1*t2-3+(2*exp(1)-1)...
*exp(-0.1*t2)),100*(1-2*exp(1)+exp(1)^2)*exp(-0.1*t3)]; t = [t1,t2,t3];

plot(n*T,y(1:length(n)),'o',t,ya,'-')
title('Example 4, T = 2 sec')
xlabel('Time (sec)')
ylabel('y(t)')
hold on

plot([23 24 25],[50 50 50],'o',[23 24 25],[45 45 45]); text(26,50,
'APPROXIMATE SOLUTION') text(26,45,
'EXACT SOLUTION') hold off

pause
```

5. Find the exponential Fourier series expansion of the pulse train shown in the figure ($T=1\text{sec}$, the period)



and the signal over one period is described by the relation

$$x(t) = \begin{cases} 2, & 0 \leq t \leq 1/4 \\ 0, & 1/4 < t < 3/4 \\ 2, & 3/4 \leq t \leq 1 \end{cases}$$

Design the approximations for 41, 21, 5 terms of the periodic signal and find the corresponding approximation rates.

6. To analyze in the exponential, trigonometric series and Fourier series of the sum of cosines the signal that in one period is expressed as

$$x(t) = \begin{cases} 1 + \frac{4t}{T}, & -T/2 < t \leq 0 \\ 1 - \frac{4t}{T}, & 0 \leq t < T/2 \end{cases}$$

for $T=1\text{sec}$, using 41 terms.

a. Plot the coefficients of $\ddot{y}_F$ and the approach signal each time.

b. Comment on the coefficients based on the symmetry of the signal.